Polynomial Regression for Turbine Power and Thermal Anomaly Prediction

Machine Learning, Assignment 1 | Sachin Singh Nain | Roll no. BT2024201 | Problems var1 and var2
Code, figures and predictions: github.com/SachinSNain/ML_Assignment-1

1. Overview

This assignment asked for polynomial regression on two problems. **var1** predicts the net power output of a steam turbine from 6 operating variables. **var2** predicts a subterranean thermal anomaly score from 3 spatial coordinates. Only polynomial models were allowed, so the real decisions were which inputs to use, how high to push the degree, and how to stop the extra terms from fitting noise. I judged every model by 5-fold cross-validation on the 1,000 training rows, never by training error.

The starting hints in the problem sheet (degree 3 on x_1 to x_3 for var1, degree 4 on x_1 alone for var2) scored a cross-validated R^2 of only 0.20 and 0.06, because both leave out inputs that matter. My final models are a degree 5 Lasso for var1 and a degree 9 Ridge for var2.

	var1: turbine power	var2: thermal anomaly
Hint baseline, CV R^2	0.2050	0.0606
Final model	Degree 5, Lasso, $\alpha = 0.0030$	Degree 9, Ridge, $\alpha = 0.0106$
CV R^2	0.9689 ± 0.0016	0.9948 ± 0.0005
CV MSE	0.3068 ± 0.0176	0.2389 ± 0.0116
Terms used	127 of 462	220 of 220

2. The data and what I expected from it

	var1: turbine power	var2: thermal anomaly
Rows, inputs	1,000 train + 1,000 test rows, 6 inputs scaled to [-1, 1]	1,000 train + 1,000 test rows, 3 inputs scaled to [-1, 1]
Target mean, std. dev.	0.7377, 3.1560	2.2902, 6.7647
Target range	-10.37 to 11.39	-30.12 to 40.54

Input	var1: what it is	var2: what it is
x_1	High-pressure steam valve adjustment	East-west offset from basecamp (m)
x_2	Condenser coolant flow rate adjustment	North-south offset from basecamp (m)
x_3	Re-injection pump hydraulic pressure	Vertical depth offset (m)
x_4	Turbine blade pitch angle	
x_5	Non-condensable gas exhaust valve rate	
x_6	Steam inlet pressure adjustment	

The physics suggested different shapes for the two models. Turbine variables do interact (inlet pressure with valve opening, for example), but a product of five different controls rarely means anything physical, so a good

high-degree model for var1 should be **sparse**. Underground temperature follows steady-state heat conduction, which is smooth in all three directions, so a good model for var2 should be **dense and smooth**: keep every term, keep them small. This expectation decided which penalty I used in section 5.

3. Method

Polynomial regression expands the d inputs into every monomial up to degree D and fits a linear model on those terms:

$$\hat{y} = b + \sum_{j=1}^{P} w_j \phi_j(\mathbf{x})$$

$$P = \binom{d+D}{D} = \frac{(d+D)!}{d!\ D!}$$

The number of terms P grows quickly. With 1,000 training rows, var1 at degree 8 already has more terms than rows:

var1 (6 inputs)	Degree 3	4	5	8	var2 (3 inputs)	Degree 4	8	9	15
Terms	84	210	462	3,003	Terms	35	165	220	816

Plain OLS fits noise once the term count gets large, so I compared two penalties added to the squared error. **Lasso** pushes unneeded weights to exactly zero, which is automatic term selection. **Ridge** shrinks every weight a little without removing any, which suits smooth, strongly correlated terms.

$$\text{Lasso}: \quad \text{Loss} = \text{MSE} + \alpha \sum_j |w_j|$$

$$\text{Ridge}: \quad \text{Loss} = \text{MSE} + \alpha \sum_j w_j^2$$

4. Experiments

4.1 Baseline with the hinted setups

Using only the hinted inputs left var1 at CV R^2 0.2050 (MSE 7.85) and var2 at 0.0606 (MSE 42.54). Dropping x_4 to x_6 on var1, and two of the three spatial axes on var2, puts a floor on the error that no choice of degree can lower.

4.2 OLS sweep over the degree, using every input

With all inputs included, I swept the degree with plain OLS. var1 improved up to degree 4, slipped at degree 5 and then broke down at degree 6, where the design matrix becomes badly conditioned. var2 improved up to degree 8 and then began to drift back up. Figure 1 below shows the whole curve, with the regularized models marked.

var1 degree	CV R^2	CV MSE
1	0.1142	8.75
2	0.7231	2.74
3	0.9081	0.91
4 (best)	0.9287	0.70
5	0.8693	1.26
6	-9.2603	100.14

var2 degree	CV R^2	CV MSE
1	0.2424	34.30
4	0.9121	3.97
6	0.9879	0.55
7	0.9931	0.31
8 (best)	0.9943	0.26
9	0.9939	0.28

4.3 Adding a penalty

On var1, degree 5 has 462 terms. Ridge on that expansion reached a CV R^2 of about 0.952 while Lasso reached 0.9689, so I kept Lasso. On var2, degree 9 has 220 terms. Lasso pruned terms across space and left small kinks in what should be a smooth field, while Ridge shrank all 220 terms evenly and reached 0.9948. I then tuned α on a fine grid for each.

	var1: Lasso	var2: Ridge
Degree / α	5 / 0.0030	9 / 0.0106
Train R^2, MSE	0.9779, 0.2203	0.9967, 0.1523
CV R^2, MSE	0.9689, 0.3068	0.9948, 0.2389
Terms	127 active, 335 (72.5%) set to zero	all 220 kept
Weight size	largest about 2.8	L2 norm 33.76, largest 9.43

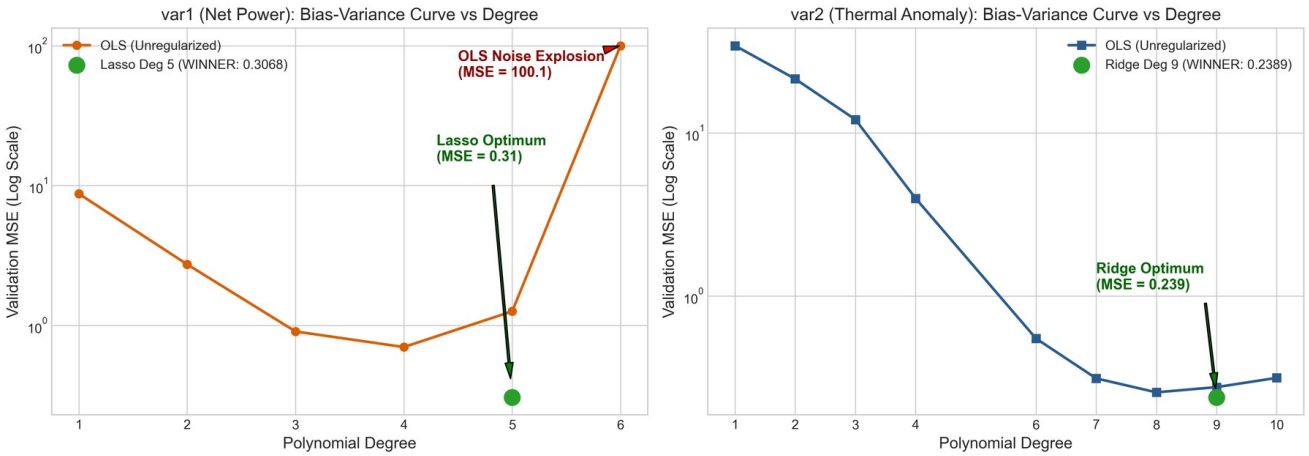


Figure 1. Validation MSE against polynomial degree (log scale). The green dots are the final regularized models, which beat the best OLS fit on both problems.

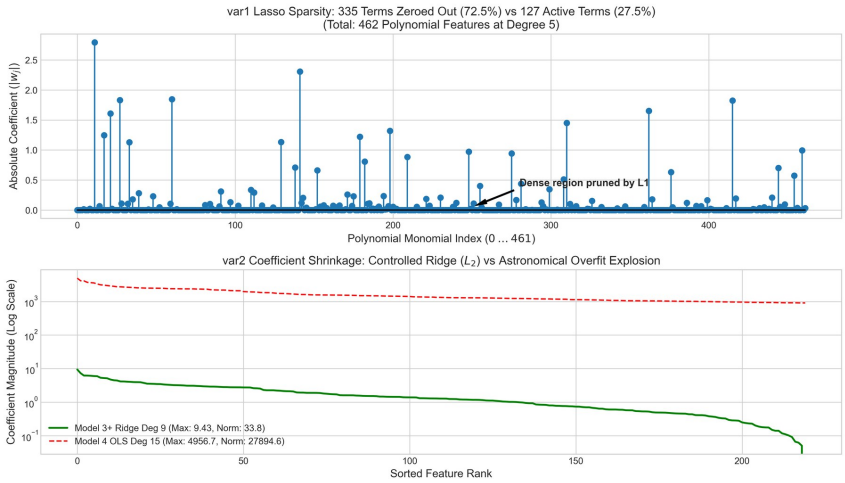


Figure 2. Top: var1 Lasso coefficients, with most of the 462 terms at zero. Bottom: sorted coefficient sizes for the var2 Ridge model and for the degree 15 OLS fit (log scale).

4.4 What overfitting looks like

To see the failure directly, I fitted plain OLS far beyond what the data supports: degree 8 on var1 (3,003 terms for 1,000 rows) and degree 15 on var2 (816 terms).

	Train R^2	CV R^2	Train MSE	CV MSE
var1, degree 8 OLS	1.0000	0.5433	0.0004	4.4843
var2, degree 15 OLS	0.9989	-0.3188	0.0485	59.8658

Training error is almost zero in both cases, but on var2 the validation score is worse than predicting the mean. The var2 weights blow up: the largest is 4,956.66 and the L2 norm is 27,894.56, against 33.76 for the Ridge model. This is the usual high-degree oscillation, and it is why the regularized models sit at the sweet spot.

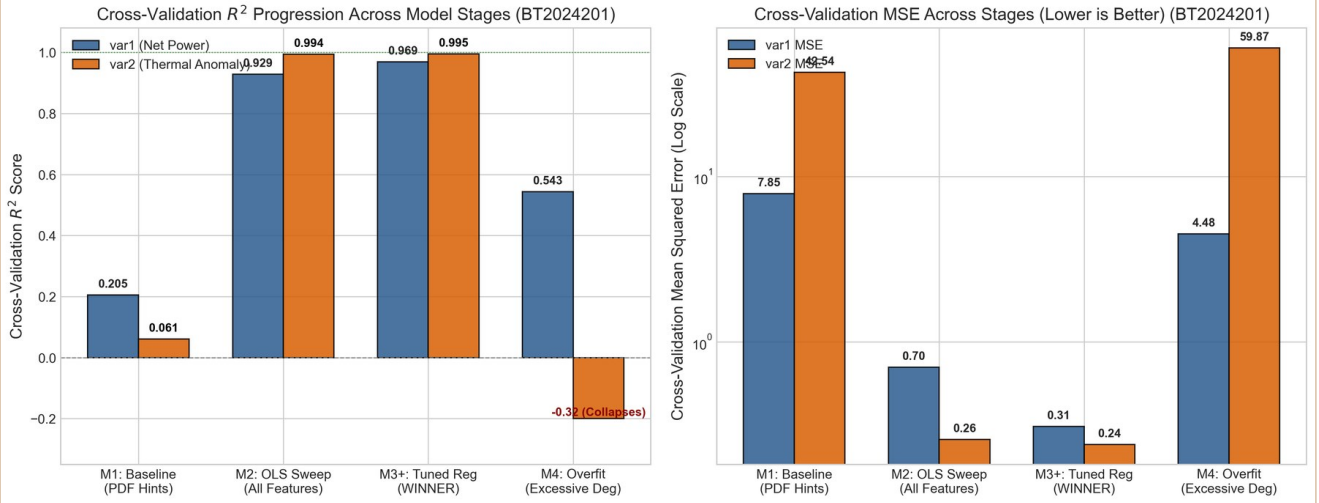


Figure 3. Cross-validation R^2 (left) and MSE (right, log scale) at each of the four stages: hint baseline, OLS sweep, tuned regularized model, overfit OLS.

5. Why I chose these models

Each choice below was made from cross-validation results, not from training error. The table lists what I compared and what I kept.

Choice	What I compared	What I kept, and why
Inputs	Hinted subsets against all inputs	All inputs. The hints dropped real signal: CV R^2 went from 0.20 to 0.93 on var1 and from 0.06 to 0.99 on var2.
Degree	OLS at degrees 1 to 9, then penalized fits at higher degrees	Degree 5 for var1 and 9 for var2. OLS peaked at 4 and 8 and became unstable above that, while a penalty let the extra terms help without adding noise.
Penalty for var1	Lasso against Ridge at degree 5	Lasso (0.9689 against about 0.952). Most fourth and fifth order products are noise, and Lasso removed 335 of the 462 terms.
Penalty for var2	Ridge against Lasso at degree 9	Ridge. A temperature field is continuous, so dropping single terms leaves artificial steps. Ridge keeps every term but holds the weights small and gave the lowest CV error.
Strength α	A fine grid for each penalty	0.0030 for Lasso and 0.0106 for Ridge, the values with the lowest cross-validated MSE.

6. Checking the final fit

Out-of-fold predictions sit on the 1:1 line for both problems, and the largest misses on var1 are about 2 units against a target range of more than 20. Residuals are centred on zero (mean 0.003 on var1, -0.010 on var2) with standard deviations of 0.554 and 0.489. They are roughly bell-shaped and show no funnel against the predictions, so the error does not grow with the size of the prediction.

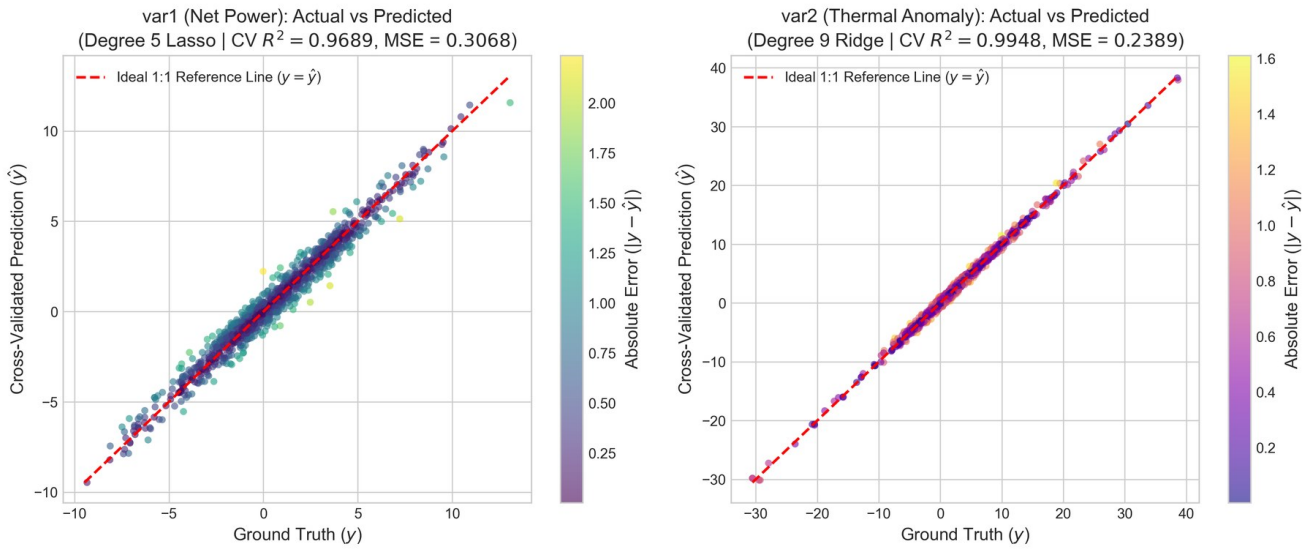


Figure 4. Cross-validated predictions against true values. Colour shows absolute error.

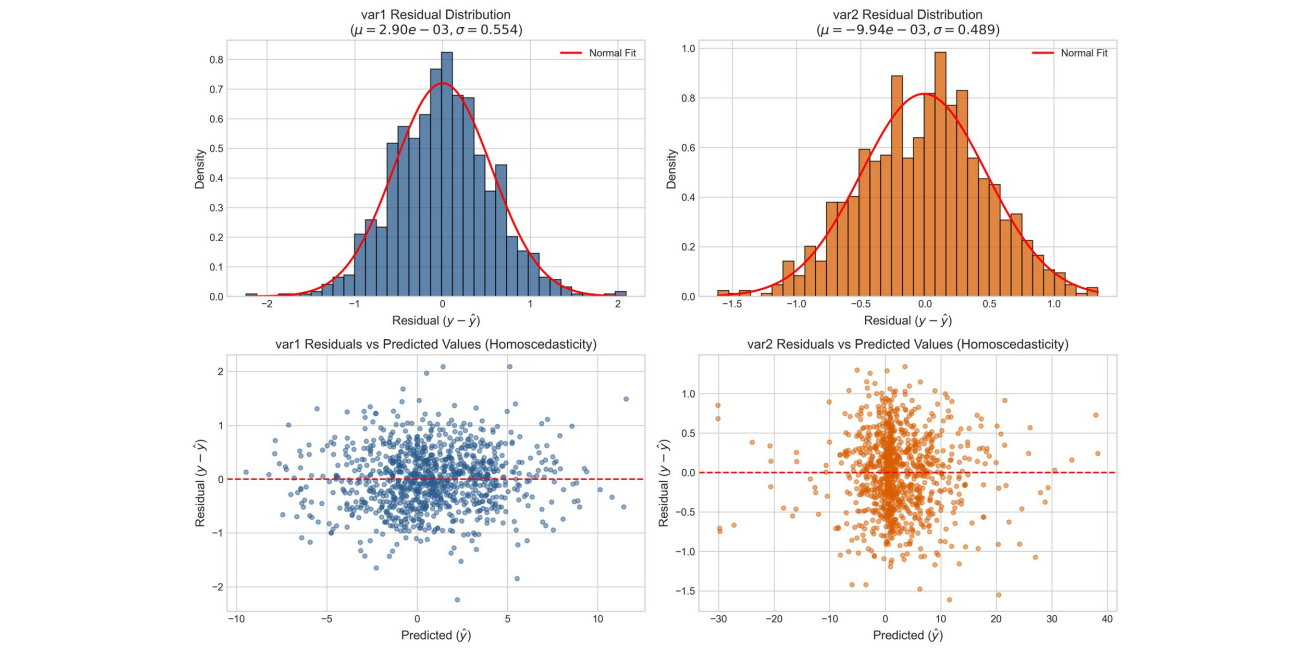


Figure 5. Residual histograms with a normal curve (top) and residuals against predictions (bottom).

7. Test predictions and conclusion

I generated predictions for both unlabelled test sets from the final models. Each file holds exactly 1,000 values under a single header y and has no index column.

File	Model	Predicted range	Training target range
`BT2024201_pred_var1.csv`	Degree 5 Lasso	-7.45 to 9.88	-10.37 to 11.39
`BT2024201_pred_var2.csv`	Degree 9 Ridge	-28.40 to 38.65	-30.12 to 40.54

All code, figures and the prediction files are in the repository: github.com/SachinSNain/ML_Assignment-1.

Both predicted ranges sit just inside the training target ranges, as expected for a smoothed fit.

Three things stood out:

- Using every input mattered more than anything else. It took CV R^2 from 0.20 to 0.93 on var1 and from 0.06 to 0.99 on var2.
- Plain OLS stops being usable at a degree set by the term count (degree 5 to 6 on var1). A penalty pushes that limit out.
- The penalty should match the problem: Lasso where most interactions are noise, Ridge where the signal is smooth and spread over every term.